Lecture 21

Question no.1

Find the first two terms in the expansion of $(4 - x)^6$, in ascending powers of x.

Solution:

$$(4-x)^6 = \binom{6}{0}(4)^6(-x)^0 + \binom{6}{1}(4)^5(-x)^1 + \binom{6}{2}(4)^4(-x)^2 + \binom{6}{3}(4)^3(-x)^3 + \binom{6}{4}(4)^2(-x)^4 + \binom{6}{5}(4)^1(-x)^5 + \binom{6}{6}(4)^0(-x)^6$$

$=4096-6144x+3840x^2+\ldots\ldots$

Check first two terms only which are 4096-6144x

Question no.2

Find the value of the constant *a* for which the coefficient of *x* in the expansion of $(1 + ax)(2 - x)^6$ is *128*.

Solution:

$(1 + ax)(2 - x)^6 = (1 + ax)(64 - 192x + 240x^2)$.

Collecting the *x* terms only, we get, $-192x + 64ax = (64a - 192)x$. Since the coefficient of *x* is given to be 64, we get $64a - 192 = 128 \rightarrow a = 5$

Question no.3

Calculate the coefficient of x^8y^3 and xy^{10} in the expansion of $(x+y)^{11}$

Solution:

11C8 = 165

11C1 = 11

Lecture 22.

Question no.1

Write the distance formula to find distance between two points $A(x_1, y_1)$ *and* $B(x_2, y_2)$.

Solution:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Question no.2

What happens if the slopes of two lines are equal?

Solution:

The two lines are parallel.

Question no.3

Find the midpoint of a line whose coordinates are A (4, 8) and B (8, 6).

Solution:
The midpoint of AB has coordinates (4+8/2 , 8+6/2) =(6,7)

Question no.4

Find the Slope of the line through the points (2, 2) and (6, 4)
Solution:
Slope = y_2-y_1/x_2-x_1 =4-2/6-2=2/4=1/2